



BrightLearn
AI & DATA SKILLS

STUDENT EXERCISE · SQL DATE
FUNCTIONS

SQL Date Functions Exercise

Practise DAYNAME, MONTH, YEAR, DATEDIFF, DATEADD,
DATE_TRUNC, LAST_DAY and more across twenty questions.
Extract, format, calculate — write the SQL.

TOPIC

SQL Date Functions

QUESTIONS

20 + 1 bonus

LEVEL

Beginner–Intermediate

INCLUDES

1 table per question



00

Welcome to the Exercise

This exercise has twenty questions on SQL date functions plus a bonus. Every question has its own table and a real business scenario. Note that date function syntax can differ between platforms — use the style your environment supports.

```
-- Snowflake / Databricks:    DAYNAME(d)  MONTHNAME(d)  MONTH(d)  YEAR(d)
-- BigQuery:                  FORMAT_DATE('%A',d)  EXTRACT(MONTH FROM d)
-- PostgreSQL:                TO_CHAR(d, 'Day')  EXTRACT(MONTH FROM d)
-- MySQL / Databricks-SQL:    DAYNAME(d)  MONTH(d)  YEAR(d)  DATEDIFF(a,b)
-- DATEADD → DATE_ADD(d, INTERVAL n DAY) or DATEADD(DAY, n, d) (platform-specific)
```

**USE THE RIGHT PLATFORM
SYNTAX**

All SQL platforms support date functions but the syntax varies. The question tells you which function concept to use — apply the correct syntax for your environment (Snowflake, Databricks, BigQuery, PostgreSQL, or MySQL). Both the function name AND the argument order may differ.



01 Questions 1–5 — Extract Date Parts

Extract individual pieces from a date column — the day name, month name, month number, year, and day of the month.

QUESTION 01

Day Name from Order Date

Table · orders

order_id	customer_id	order_date
1001	101	2026-05-01
1002	102	2026-05-02
1003	103	2026-05-03
1004	104	2026-05-04
1005	105	2026-05-05

TASK

Return each order with the day name of the order date.

USE

DAYNAME · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, day_name

WRITE YOUR SQL HERE

QUESTION 02

Month Name from Signup Date

Table · customer_signups



customer_id	customer_name	signup_date
201	John	2026-01-15
202	Mary	2026-02-20
203	Peter	2026-03-05
204	Sarah	2026-04-18
205	Thabo	2026-05-30

T A S K

Return each customer with the name of the month in which they signed up.

U S E

MONTHNAME · **AS**

EXPECTED OUTPUT COLUMNS

customer_id, customer_name, signup_date, signup_month_name

WRITE YOUR SQL HERE

QUESTION 03**Month Number from Sale Date**

Table · sales

sale_id	product_name	sale_date	amount
S001	Laptop	2026-01-10	12000
S002	Mouse	2026-02-15	250
S003	Keyboard	2026-03-20	700
S004	Monitor	2026-04-25	3500
S005	Desk	2026-05-30	2000

T A S K

Return each sale with the month number extracted from sale_date.



USE

MONTH · AS

EXPECTED OUTPUT COLUMNS

sale_id, product_name, sale_date, sale_month

WRITE YOUR SQL HERE

QUESTION 04

Year from Transaction Date

Table · transactions

transaction_id	customer_id	transaction_date	amount
T001	101	2024-12-15	500
T002	102	2025-01-20	1200
T003	103	2025-06-10	800
T004	104	2026-02-05	1500
T005	105	2026-05-25	2000

TASK

Return each transaction with the year extracted from transaction_date.

USE

YEAR · AS

EXPECTED OUTPUT COLUMNS

transaction_id, customer_id, transaction_date, transaction_year

WRITE YOUR SQL HERE



QUESTION 05

Day of Month from Delivery Date

Table · deliveries

delivery_id	customer_id	delivery_date
D001	101	2026-05-01
D002	102	2026-05-08
D003	103	2026-05-15
D004	104	2026-05-22
D005	105	2026-05-29

TASK

Return each delivery with the day of the month extracted from delivery_date.

USE

DAY · AS

EXPECTED OUTPUT COLUMNS

delivery_id, customer_id, delivery_date, day_of_month

WRITE YOUR SQL HERE



02

Questions 6–10 — Current Date, Convert & Format

Use the current date, convert text strings to date values, and format dates for display.

QUESTION 06

Add Today's Date

Table · employees

employee_id	employee_name	department
301	Nandi	Sales
302	Brian	IT
303	Lerato	Finance
304	Sipho	HR
305	Aisha	Marketing

TASK

Return all employees and add a column showing today's date.

USE

CURRENT_DATE() · AS

EXPECTED OUTPUT COLUMNS

employee_id, employee_name, department, today_date

WRITE YOUR SQL HERE

QUESTION 07

Convert Text to Date

**Table · online_orders**

order_id	customer_id	order_date_text
4001	101	2026-01-15
4002	102	2026-02-20
4003	103	2026-03-25
4004	104	2026-04-10
4005	105	2026-05-05

T A S K

Convert the order_date_text string column into a proper date column.

U S E

TO_DATE · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date_text, order_date

WRITE YOUR SQL HERE

QUESTION 08**Format Date as Text****Table · payment_dates**

payment_id	customer_id	payment_date
P001	101	2026-01-05
P002	102	2026-02-10
P003	103	2026-03-15
P004	104	2026-04-20
P005	105	2026-05-25

T A S K



Format payment_date as a text string in the format YYYY-MM-DD.

USE

TO_CHAR · **AS**

EXPECTED OUTPUT COLUMNS

payment_id, customer_id, payment_date, formatted_payment_date

WRITE YOUR SQL HERE

QUESTION 09

Days Since Last Purchase

Table · customer_purchases

customer_id	customer_name	last_purchase_date
501	John	2026-05-01
502	Mary	2026-05-10
503	Peter	2026-05-15
504	Sarah	2026-05-20
505	Thabo	2026-05-25

TASK

Calculate how many days have passed since each customer's last purchase. Use today's date.

USE

CURRENT_DATE() · **DATEDIFF** · **AS**

EXPECTED OUTPUT COLUMNS

customer_id, customer_name, last_purchase_date, days_since_last_purchase

WRITE YOUR SQL HERE



QUESTION 10

Expected Delivery Date

Table · shipping_orders

order_id	customer_id	order_date
6001	101	2026-05-01
6002	102	2026-05-03
6003	103	2026-05-05
6004	104	2026-05-07
6005	105	2026-05-09

TASK

Calculate the expected delivery date by adding 7 days to the order_date.

USE

DATEADD · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, expected_delivery_date

WRITE YOUR SQL HERE



03

Questions 11–15 — Multi-Extract & Date Filtering

Extract multiple date parts at once, filter rows by year or month, and use `LAST_DAY` and `DATE_TRUNC`.

QUESTION 11

Year, Month, Day Separately

Table · bookings

booking_id	customer_id	booking_date
B001	101	2026-01-12
B002	102	2026-02-18
B003	103	2026-03-22
B004	104	2026-04-09
B005	105	2026-05-27

TASK

Extract the year, month number, and day from `booking_date` into three separate columns.

USE

`YEAR` · `MONTH` · `DAY` · `AS`

EXPECTED OUTPUT COLUMNS

`booking_id`, `customer_id`, `booking_date`, `booking_year`, `booking_month`, `booking_day`

WRITE YOUR SQL HERE

QUESTION 12

**Filter Orders from 2026****Table · yearly_orders**

order_id	customer_id	order_date	amount
7001	101	2024-12-15	500
7002	102	2025-01-20	1200
7003	103	2025-06-10	800
7004	104	2026-02-05	1500
7005	105	2026-05-25	2000

T A S K

Return only orders from the year 2026.

U S E

YEAR · WHERE · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, order_year, amount

WRITE YOUR SQL HERE

QUESTION 13**Filter Orders from March****Table · monthly_orders**

order_id	customer_id	order_date	amount
8001	101	2026-01-15	500
8002	102	2026-02-20	1200
8003	103	2026-03-10	800
8004	104	2026-03-25	1500
8005	105	2026-05-30	2000

**T A S K**

Return only orders placed in March (month number 3).

U S E

MONTH · **WHERE** · **AS**

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, order_month, amount

WRITE YOUR SQL HERE

QUESTION 14**Last Day of the Month**

Table · subscriptions

subscription_id	customer_id	start_date
SUB001	101	2026-01-10
SUB002	102	2026-02-15
SUB003	103	2026-03-20
SUB004	104	2026-04-25
SUB005	105	2026-05-30

T A S K

Return the last day of the month for each subscription start date.

U S E

LAST_DAY · **AS**

EXPECTED OUTPUT COLUMNS

subscription_id, customer_id, start_date, month_end_date

WRITE YOUR SQL HERE



QUESTION 15

First Day of the Month

Table · campaign_sends

send_id	customer_id	send_date
C001	101	2026-01-12
C002	102	2026-02-18
C003	103	2026-03-22
C004	104	2026-04-09
C005	105	2026-05-27

TASK

Return the first day of the month for each campaign send date.

USE

DATE_TRUNC · **AS**

EXPECTED OUTPUT COLUMNS

send_id, customer_id, send_date, month_start_date

WRITE YOUR SQL HERE



04 Questions 16–20 — Advanced Date Work

The final five questions involve formatting, age calculations, CASE with dates, quarter extraction, and filtering by date difference.

QUESTION 16

Format as Month and Year

Table · `invoice_dates`

invoice_id	customer_id	invoice_date
INV001	101	2026-01-05
INV002	102	2026-02-10
INV003	103	2026-03-15
INV004	104	2026-04-20
INV005	105	2026-05-25

TASK

Format `invoice_date` as a text string showing month name and year. Example: January 2026.

USE

`TO_CHAR` · `AS`

EXPECTED OUTPUT COLUMNS

`invoice_id`, `customer_id`, `invoice_date`, `invoice_month_year`

WRITE YOUR SQL HERE

QUESTION 17

Customer Age from Date of Birth

**Table · customer_birthdays**

customer_id	customer_name	date_of_birth
901	John	1998-05-10
902	Mary	1990-08-20
903	Peter	2002-03-15
904	Sarah	1985-12-01
905	Thabo	2000-07-30

T A S K

Calculate each customer's age in years using their date of birth and today's date.

U S E

CURRENT_DATE() · **DATEDIFF** or **YEAR** · **AS**

EXPECTED OUTPUT COLUMNS

customer_id, customer_name, date_of_birth, customer_age

WRITE YOUR SQL HERE

QUESTION 18**Classify Weekend vs Weekday****Table · weekend_orders**

order_id	customer_id	order_date
9001	101	2026-05-01
9002	102	2026-05-02
9003	103	2026-05-03
9004	104	2026-05-04
9005	105	2026-05-05

T A S K



Classify each order as Weekend or Weekday based on the day name of the order date.
Saturday and Sunday = Weekend. All others = Weekday.

U S E

DAYNAME · CASE · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, day_name, day_type

W R I T E Y O U R S Q L H E R E

Q U E S T I O N 1 9

Transactions by Quarter

Table · quarterly_transactions

transaction_id	customer_id	transaction_date	amount
Q001	101	2026-01-15	500
Q002	102	2026-03-20	1200
Q003	103	2026-04-10	800
Q004	104	2026-07-05	1500
Q005	105	2026-10-25	2000

T A S K

Extract the quarter number from each transaction date.

U S E

QUARTER · AS

EXPECTED OUTPUT COLUMNS

transaction_id, customer_id, transaction_date, transaction_quarter, amount

W R I T E Y O U R S Q L H E R E



QUESTION 20

Orders Older Than 30 Days

Table · recent_orders

order_id	customer_id	order_date	amount
R001	101	2026-04-01	500
R002	102	2026-04-15	1200
R003	103	2026-05-01	800
R004	104	2026-05-10	1500
R005	105	2026-05-25	2000

TASK

Return only orders that are more than 30 days old from today. Calculate days_since_order using today's date.

USE

CURRENT_DATE() · DATEDIFF · WHERE · AS

EXPECTED OUTPUT COLUMNS

order_id, customer_id, order_date, days_since_order, amount

WRITE YOUR SQL HERE





05

Bonus Challenge

The bonus combines DATEDIFF, CASE, and CURRENT_DATE into a single customer segmentation query. Apply every concept from this guide in one question.

QUESTION BONUS

Customer Recency Segment

Table · customer_recency

customer_id	customer_name	last_purchase_date	total_spend
1001	John	2026-05-25	5000
1002	Mary	2026-05-10	2500
1003	Peter	2026-04-01	700
1004	Sarah	2026-02-15	15000
1005	Thabo	2025-12-20	300

TASK

Calculate days_since_last_purchase for each customer using CURRENT_DATE. Then use CASE to classify: ≤ 30 days = Active Customer, 31–90 days = At Risk Customer, > 90 days = Inactive Customer.

USE

CURRENT_DATE() · DATEDIFF · CASE · AS

EXPECTED OUTPUT COLUMNS

customer_id, customer_name, last_purchase_date, days_since_last_purchase, customer_status

WRITE YOUR SQL HERE



Extract it. Calculate it.

Master SQL date functions by writing real queries on
twenty business tables — from extracting day names
to calculating customer recency segments.

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